

Data Quality and Evaluation Design in Graph-Based Dengue Forecasting

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Abstract—Weekly dengue forecasting requires comparisons that separate model skill from data and evaluation artifacts. We audit a Sri Lankan graph-forecasting benchmark and reanalyze stored graph and susceptible–exposed–infectious–recovered (SEIR) model results against persistence. The study covers a legacy array and a rebuilt series of 559 reporting weeks for 25 districts, with seven missing reports retained as missing. An inconsistent published source-table row produces a national district sum of 7165 cases. Excluding forecast windows touching that row lowers the mean persistence root mean squared error from 44.80 to 29.52 cases per district-week. This is a scoring-sensitivity diagnostic, not a corrected-model experiment. The historical rolling-origin implementation shares two target weeks at each training/validation and validation/evaluation boundary. Its stored scores therefore support retrospective development analysis rather than independent confirmation. Across nine origins, persistence scores 28.54 and an anchored adaptive SEIR variant scores 27.45; the paired difference is -1.10 with a descriptive 95% interval of [-2.84, 0.65]. None of the 30 learned-model comparisons is significant under the protocol-specified Holm correction. The audit also identifies a mismatch between specified and implemented multiplicity corrections. These findings show why source integrity, target-level split checks, and strong simple baselines must precede claims of forecasting improvement. They do not establish model equivalence or an isolated benefit from SEIR equations.

Index Terms—data quality, dengue forecasting, evaluation design, graph neural networks, persistence baseline, SEIR model

I. INTRODUCTION

Dengue causes recurring epidemics in Sri Lanka [1]. District-level forecasts could support clinical preparation and vector control. Graph neural networks represent spatial relationships, while epidemic compartment models encode assumptions about disease progression. Both are plausible forecasting tools, but their value depends on the data representation and comparison design [2], [3].

Persistence, which repeats the latest reported count, is an essential first reference for short-horizon forecasts. A complex model’s low error alone does not show that it adds useful predictive information. The benchmark studied here compares graph encoders with statistical and neural alternatives, but does not report persistence [2]. Our audit identifies errors in its reused data representation and limitations in our subsequent model evaluation. These observations motivate the question:

how do data quality, persistence baselines, and evaluation design affect conclusions about graph-based dengue forecasting?

We answer this as a methodological audit, not as validation of a new forecasting model. The completed evidence comprises source-report consistency checks, an exclusion-based sensitivity analysis, inspection of implemented splits and graphs, and reanalysis of saved development scores. No untouched-period result is reported. Our contributions are:

- 1) A source-table audit and reproducible measurement of how affected forecast windows change persistence error.
- 2) An evaluation audit identifying shared target weeks, erroneous geographic entries, and a discrepancy in multiplicity correction.
- 3) A retrospective comparison of graph and SEIR heads with persistence, reporting paired uncertainty and the limits of mechanistic attribution.

II. RELATED WORK

Weng et al. [2] adapt graph encoders for Sri Lankan dengue forecasting. They compare STGAT, A3T-GCN, AST-GCN, DCRNN, and AAGCN, which respectively use graph attention, temporal attention, spatial-temporal attention, diffusion recurrence, and adaptive graph convolution [4]–[8]. Our STGAT follows the dengue adaptation, rather than the original pedestrian-trajectory formulation. DengueGNN also studies graph-based dengue prediction [9]. Graph WaveNet provides the learned-adjacency construction used in our adaptive encoder [10].

Liu et al. [3] use an LSTM to predict a force of infection for a human SEIR model in Sri Lanka. EINNs and CausalGNN incorporate epidemiological structure into neural forecasting, while MepoGNN couples graph learning with metapopulation dynamics [11]–[13]. These studies motivate mechanistic structure, but do not establish its value under our data, loss, and forecast horizon. Our contribution concerns the evidence needed to interpret such comparisons.

III. DATA AND AUDIT PROCEDURE

A. Data representations and source checks

The rebuilt series contains 559 reporting weeks for 25 districts, from June 15, 2013, to February 24, 2024; seven reports are missing. Missing reports remain missing, and

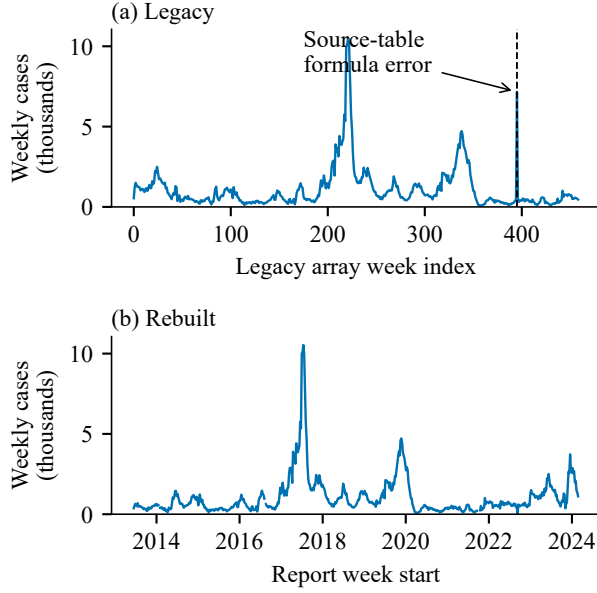


Fig. 1. National district sums in the legacy array and rebuilt series. The legacy horizontal axis is array position, not a verified calendar. The marked row contains a source-table error. Missing reports interrupt the rebuilt curve; correcting plot dates does not update historical model scores.

windows touching missing case inputs or targets are excluded. These dates use the repaired report calendar rather than ISO week labels. The task uses three past weeks to forecast the next three weeks. The earlier benchmark representation, called the legacy array here, contains 459 rows. Fig. 1 distinguishes the two representations; they do not define interchangeable evaluation periods.

We compare district sums with printed national totals and examine the within-row arithmetic in the original report. Legacy row 395 corresponds to the report covering December 26, 2020, to January 1, 2021 [14]. In weekly row A, 15 of 24 inner cells equal the sum of the preceding two cells; its district sum is 7165, while its national cell is 35. The repeated additive pattern is consistent with an erroneous spreadsheet formula propagated across the row. Cumulative row B, including Kalmunai, sums to its printed national total of 351. For this first reporting week of the year, the cumulative row supplies the replacement weekly vector. This is a source-table inconsistency, not evidence of a genuine outbreak or a reporting backlog.

B. Sensitivity diagnostic and provenance

We reproduce persistence on the legacy protocol, then exclude every evaluation window whose input or target touches the erroneous row. The protocol averages RMSE across three origins, with 68 test starts per origin before exclusion. The mask removes six starts from the last origin and changes neither the split boundaries nor the remaining forecasts. This diagnostic measures sensitivity to the affected windows; it is not a comparison of models retrained after source correction. The rebuilt-series model comparison is reported separately.

We also compare the stored adjacency with a border list derived from GADM polygons. The corrected graph contains 57 shared borders, represented as 114 directed neighbor entries; the loader adds 25 self-loops, giving 139 entries. The historical file has 141 entries after self-loops because it additionally lists Kandy to Ampara and Kegalle to Kalutara. These are two one-directional non-border entries, not two valid shared borders. The corrected file passes the stored border and symmetry checks. The historical model results below use the earlier graph.

IV. MODELS IN THE HISTORICAL COMPARISON

The models serve as case studies in the audit. Fig. 2 separates the graph representation, persistence anchor, and optional mechanistic decoder. Stored nine-origin encoders use case histories only; SEIR heads also use population and initial compartment states.

A. Representations and output heads

Counts $c_{i,t}$ are transformed using training-period statistics,

$$z_{i,t} = \{\log(1 + c_{i,t}) - \mu\} / \sigma. \quad (1)$$

The persistence anchor p repeats the standardized latest count for every horizon. Our GCN projects the input to 64 dimensions and uses two residual graph layers, $H' = H + \text{Dropout}(\text{ReLU}(AHW + b))$. Static A is row-normalized adjacency with self-loops. The adaptive version uses $A_{\text{ad}} = \text{softmax}(\text{ReLU}(E_1 E_2^\top))$, with standard-normal 25×10 embeddings shared across layers and exempt from weight decay. The exemption was not independently ablated. Published comparison encoders replace this representation stage.

For read-out r , direct, residual, and gated non-SEIR heads output r , $p + r$, and $p + g(r - p)$ respectively. The gate is $g = \text{sigmoid}(a)$ with a initialized to -2 . A bare SEIR head outputs the transformed simulator count $z(\hat{c})$; its gated version outputs $p + g\{z(\hat{c}) - p\}$. All are converted back to nonnegative counts for scoring. Squared error is applied in standardized log space, so exponentiating its conditional-mean predictor need not minimize count-scale squared error.

B. SEIR decoder and assumptions

The decoder advances susceptible, exposed, infectious, and recovered fractions with daily exponential flows:

$$\begin{aligned} f_S &= S(1 - e^{-\lambda \Delta t}), & f_E &= E(1 - e^{-\omega \Delta t}), \\ f_I &= I(1 - e^{-\gamma \Delta t}), & \hat{c}_{i,h} &= \rho_{\text{eff}} N_i \sum_{d=1}^7 f_E^{(d)}. \end{aligned} \quad (2)$$

Flows are computed from the previous state, then transferred along $S \rightarrow E \rightarrow I \rightarrow R$. Weekly incidence sums exposed-to-infectious flows and scales by population N_i and the effective reporting fraction.

The implementation fixes $\omega = 0.7/7 = 0.1$ and $\gamma = 1/7$ per day, using seven daily steps. These effective SEIR rates follow Liu et al. [3] and correspond to mean exposed and infectious durations of ten and seven days, rather than district-specific estimates. The base reporting fraction $\rho = 1/11$

Historical forecasting paths

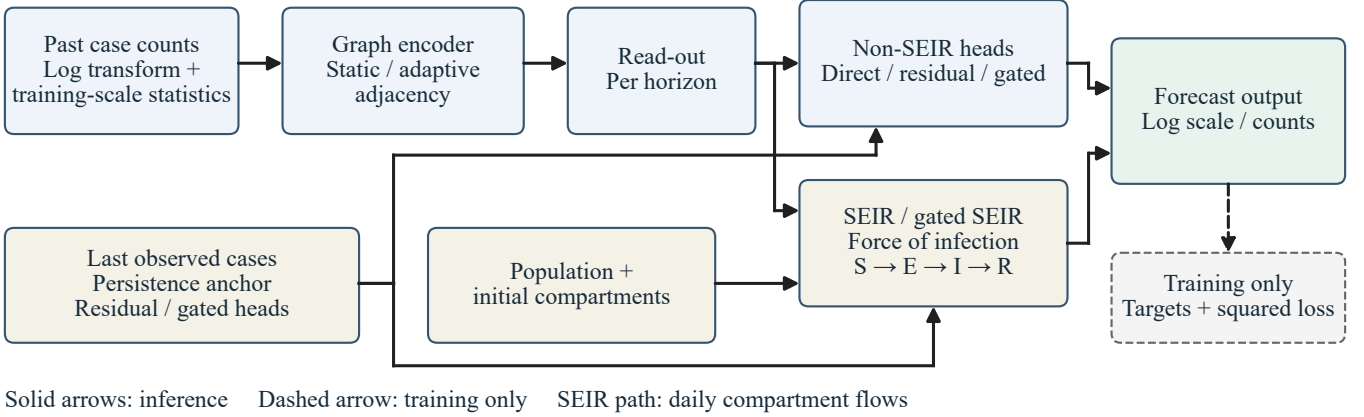


Fig. 2. Historical model family. The encoder/read-out feeds either a non-SEIR head or a force-of-infection head with daily SEIR flows. Persistence supplies the residual or gated anchor. Population and initialized compartments enter the SEIR branch. Count forecasts are evaluated after inverse transformation; the dashed loss path is used only in training. Alternative paths describe separate model variants.

follows the under-ascertainment scale used in the motivating compartmental model [3]. When state fitting is enabled, a learned scalar q gives $\rho_{\text{eff}} = \rho \exp(q)$; this scale is not independently identified as a surveillance reporting probability.

Initial infectious and exposed fractions use $c_{t-1}/(\rho N_i)$ and $c_{t-2}/(\rho N_i)$, clipped to $[10^{-6}, 0.5]$. The susceptible fraction is $\text{clip}(0.318 - C_{<t}/(\rho N_i), 0.01, 1)$, where $C_{<t}$ sums available earlier reports, with missing contributions set to zero in this state calculation. The initial 0.318 complements a reported 68.2% seroprevalence in a suburban Colombo cohort [15]; it is a modeling assumption, not a national susceptibility estimate. A learned multiplier $2 \text{sigmoid}(v)$ modifies the exposed fraction, globally or from the encoder; recovered mass compensates subject to clipping. The main nine-origin comparison does not vary the fixed progression rates or initial susceptibility, so it supplies no sensitivity evidence for transferring them across districts or years.

Free-rate heads exponentiate a clipped read-out plus learned bias. Anchored heads supply the population and susceptible scale explicitly,

$$\lambda_{i,h}^* = \frac{\max(c_{i,t-1}, 0.5)}{7\rho_{\text{eff}} N_i S_i} \exp\{\text{clip}(r_{i,h}, -10, 10)\}. \quad (3)$$

This approximates the rate sustaining the latest count, rather than guaranteeing persistence at zero read-out. A learned spatial import term multiplies both rates by $\exp\{0.1\beta(u_i - \bar{u})\}$, where $u_i = \log \max((AI)_i, 10^{-9})$ and $\bar{u}$ is the within-window district mean. This uses the fixed graph even with an adaptive encoder. The mass-action variant instead multiplies a positive weekly transmission coefficient by the daily updated infectious fraction and omits this import multiplier.

V. EVALUATION AND STATISTICAL AUDIT

A. Historical split and training

The nine origins are 0.50 to 0.90 in steps of 0.05 of the forecast-start list; each evaluation block covers the next 5%.

The 30 starts immediately before each cut form validation. Starts touching missing reports are dropped after setting boundaries. We use seeds 0, 1, and 2, giving 27 runs per learned arm; persistence is deterministic. The separate three-origin ablation uses origins 0.55, 0.70, and 0.85 with 15% evaluation blocks and is retained in the supplement. Scores from these protocols are not pooled.

The saved runs use Adam, learning rate 0.003, weight decay 0.0001, batch size 32, cosine decay, gradient clipping at 5, and a maximum of 300 epochs. Training minibatches are shuffled within the training partition. Early stopping restores the lowest count-scale validation RMSE, with patience 40 and no minimum-epoch requirement; improvements must exceed 10^{-6} , otherwise the earlier checkpoint is retained. These are shared fixed settings, not an equal-budget hyperparameter search. Later arms were selected after earlier evaluation scores were visible, so the experiment is retrospective development evidence.

B. Metrics and inference

We average seed scores within each origin before comparing models with persistence. Overall RMSE and MAE are means of origin-level scores in cases per district-week; horizon RMSE is computed separately. Paired differences use the same origins, and skill is the mean of $1 - \text{RMSE}_{m,o}/\text{RMSE}_{p,o}$. Within-origin seed standard deviations are reported separately in the supplement.

The exact two-sided sign-flip test enumerates all 2^9 sign assignments of the nine origin differences. The documented protocol specifies Holm correction across the 30 learned arms sharing persistence; historical software instead applied Benjamini–Hochberg (BH). We recompute Holm without changing scores, pairing, or the comparison family, and retain both corrections in the reproducibility files. Descriptive 95% t intervals use the seed-averaged origin differences. They are not simultaneous selection-adjusted intervals. Serial dependence,

Observed: historical rolling origins



Forecast starts are chronological.
Two target weeks overlap at each boundary.

Recommended after correcting the splits



Recommendation only; no confirmation result reported.

Fig. 3. Observed historical design and recommended evaluation sequence. The upper row partitions forecast starts, but adjacent target windows overlap. The lower row is a recommendation for a corrected study, not a completed confirmation. Model choice, preprocessing, and tuning must be frozen before accessing the untouched period.

overlapping targets, and retrospective choices limit inferential interpretation under either correction.

VI. RESULTS

A. Source errors change the baseline reference

On the legacy array, persistence has mean RMSE 44.80; excluding windows touching the inconsistent row reduces it to 29.52, a 34.1% reduction. This change comes from removing six starts in the final fold; that fold’s RMSE falls from 68.62 to 22.80. A large error contribution from a source artifact can therefore change the apparent difficulty of the task. It does not quantify how any learned model would perform after retraining on the repaired series. Comparisons with earlier experiments require the same input/target mask and protocol for every model and baseline.

B. Chronology alone does not prevent target overlap

Explicit target-set intersections find two shared weeks at every training/validation and validation/evaluation boundary across all nine origins. Partitioning forecast starts chronologically does not separate the multiweek outcomes. Early stopping can therefore use outcomes present in the first evaluation forecasts. This establishes a design defect; its effect on model rankings has not been measured by an embargoed rerun. A target-level exclusion rule must be checked directly rather than inferred from non-overlapping start indices.

C. Historical models remain close to persistence

Table I reports selected development arms, with inference corrected over the full stored family. Persistence has RMSE 28.54 and MAE 13.94; the GCN residual baseline has RMSE 28.12 and MAE 13.62. The latter’s paired difference is -0.42, with a descriptive interval [-1.54, 0.70]. The adaptive anchored gated-SEIR variant with learned exposed initialization scores 27.45, with difference -1.10 and interval [-2.84, 0.65]. Its raw sign-flip value is 0.199, historical BH value 0.576, and Holm value 1.000. No learned arm improves significantly under

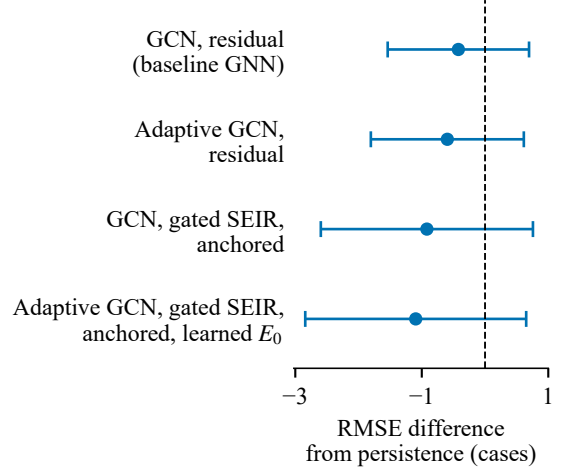


Fig. 4. Historical paired RMSE differences from persistence, with descriptive 95% t intervals across nine seed-averaged origins. Negative values favor the learned model; the dashed zero line means no difference. These intervals do not correct the split defect or retrospective selection.

either correction; the eight historical BH degradations also fail to reach significance under Holm. This does not imply equivalence or demonstrate that every architecture has been adequately tuned.

Persistence error rises from 23.47 at the first horizon to 33.26 at the third; the GCN residual baseline scores 23.24 and 32.47 respectively. Thus its small aggregate difference is not a large improvement hidden at one endpoint. The adaptive residual encoder scores 27.94, while gated adaptive SEIR variants score about 27.45. That contrast changes the head as well as its anchoring behavior and does not isolate SEIR dynamics. The supplementary three-origin matched gated comparisons have mixed validation differences and lower SEIR evaluation errors, but lack equal validation-only tuning and a matched adaptive-encoder control.

VII. DISCUSSION AND LIMITATIONS

A. Interpretation of the completed audit

Persistence is competitive in these stored comparisons. A plausible explanation is that a short forecast horizon preserves much of the current local case level; this is an interpretation, not a bound on achievable skill. Persistence also avoids estimating parameters from a small number of epidemic regimes. Its competitiveness justifies including it before interpreting improvements over more complex neural references.

The SEIR decoder introduces population scaling and a constrained progression through compartments. A persistence anchor can make this decoder easier to use without establishing that the equations themselves improve forecasts. The missing matched adaptive gated control prevents that attribution here. Likewise, learned adjacency weights are predictive parameters, not validated disease-transmission pathways. A lower error under one parameterization does not independently validate its epidemiological assumptions.

TABLE I

HISTORICAL DEVELOPMENT COMPARISON ON THE REBUILT SERIES, USING THE ORIGINAL GRAPH. RMSE AND MAE ARE IN CASES PER DISTRICT-WEEK, AVERAGED OVER NINE ORIGINS AND THREE SEEDS. HORIZON COLUMNS REPORT RMSE. Δ IS THE SEED-AVERAGED PAIRED RMSE DIFFERENCE FROM PERSISTENCE; BRACKETS GIVE A DESCRIPTIVE 95% T INTERVAL OVER ORIGINS. p_{adj} USES THE UNCHANGED EXACT SIGN-FLIP TEST WITH HOLM ADJUSTMENT OVER THIRTY ARMS. RETROSPECTIVE SELECTION AND TEMPORAL DEPENDENCE LIMIT INFERENCE. NO COMPARISON IS SIGNIFICANT UNDER HOLM. HISTORICAL BH VALUES ARE ARCHIVED SEPARATELY.

Model	$h = 1$	$h = 2$	$h = 3$	RMSE	MAE	Δ [95% CI]; p_{adj}
Persistence (last week)	23.47	27.94	33.26	28.54	13.94	–
GCN, direct	23.10	27.82	32.68	28.17	13.66	-0.37 [-1.88, +1.14]; 1.000
GCN, residual (baseline GNN)	23.24	27.79	32.47	28.12	13.62	-0.42 [-1.54, +0.70]; 1.000
LSTM, direct	23.65	28.22	33.38	28.73	14.00	+0.19 [-1.99, +2.37]; 1.000
ASTGCN, direct	23.92	28.54	33.45	28.94	14.27	+0.40 [-2.02, +2.81]; 1.000
AAGCN, direct	24.11	29.10	33.66	29.26	14.06	+0.72 [-1.57, +3.00]; 1.000
STGAT, direct	42.60	44.55	46.75	44.71	22.17	+16.17 [+4.24, +28.11]; 0.227
ASTGCN, residual	23.07	28.08	33.66	28.65	13.98	+0.11 [-1.68, +1.90]; 1.000
Adaptive GCN, residual	23.07	27.78	32.15	27.94	13.61	-0.60 [-1.81, +0.61]; 1.000
GCN, SEIR, free rate	27.58	40.81	50.14	40.77	17.97	+12.23 [+2.67, +21.80]; 0.227
GCN, gated SEIR, free rate	22.86	28.04	33.29	28.43	13.51	-0.11 [-3.23, +3.00]; 1.000
Adaptive GCN, SEIR, anchored	23.99	28.81	35.29	29.80	13.84	+1.26 [-1.47, +3.98]; 1.000
Adaptive GCN, gated SEIR, mass-action	22.90	28.02	33.19	28.39	13.52	-0.16 [-3.16, +2.85]; 1.000
GCN, gated SEIR, anchored	22.49	27.38	32.04	27.62	13.39	-0.92 [-2.60, +0.76]; 1.000
Adaptive GCN, gated SEIR, anchored	22.51	27.09	31.85	27.45	13.31	-1.09 [-2.81, +0.63]; 1.000
Adaptive GCN, gated SEIR, anchored, learned E_0	22.45	27.07	31.88	27.45	13.29	-1.10 [-2.84, +0.65]; 1.000

The source error and split defect have different implications. The former has a directly measured effect on the persistence diagnostic. The latter invalidates a strictly held-out interpretation, but its numerical influence is unknown. Neither finding licenses a claim that all earlier model gains were caused by those defects. The mismatch between specified and implemented multiplicity correction illustrates why a protocol must be checked against executed analysis code.

B. Scope and unavailable evidence

The available evidence includes neither a final model comparison using the corrected graph nor an untouched-period confirmation. Equal validation-only tuning, corrected-protocol classical baselines, and the complete matched head comparison are unavailable. Archived seasonal-naive and regression results use a different legacy protocol and cannot fill those gaps. Date-dependent climate and policy results are excluded because their input calendar was subsequently repaired. Stored population and availability rules also differ from later proposals; the historical scores are not relabeled as using the new rules.

Fixed progression rates, approximate state initialization, uncertain reporting, and susceptibility transferred from a local serosurvey limit biological interpretation. Cumulative depletion does not explicitly represent dengue serotypes, waning protection, or changing surveillance. Exploratory parameter diagnostics in other protocols do not constitute sensitivity analysis for the nine-origin comparison. The study concerns one country, one disease, and a three-week horizon.

A corrected evaluation should freeze preprocessing, model list, tuning budget, tie-breaking, target embargo, and paired testing before accessing the new period. It should include seasonal naive, autoregressive and count-regression baselines alongside persistence, and compare gated heads with and with-

out SEIR under matched settings. These are recommendations derived from the audit, not completed contributions.

VIII. CONCLUSION

The audit shows that a source-table inconsistency materially changes the persistence reference and that chronological forecast starts can conceal shared target weeks. Reanalysis also exposes a discrepancy between the stated and implemented multiplicity correction. The historical graph and SEIR comparisons do not establish an improvement over persistence or an isolated benefit from epidemic equations. Source validation, strong simple baselines, and checks of executed evaluation code are therefore necessary to interpret this benchmark’s forecasting claims.

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